

Does earnings information predict abnormal returns?

A null result on 3,323 S&P 500 announcements, and the machinery built to make that answer trustworthy.

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ABSTRACT

Post-earnings-announcement drift is among the most-studied anomalies in finance and among the easiest to reproduce spuriously. This note tests whether information in earnings releases predicts short-horizon abnormal returns across 3,323 announcements by 114 S&P 500 companies, timestamped to the minute from SEC filings, with each year from 2019 to 2024 held out and predicted by a model frozen before it began. **It does not.** Mean out-of-sample rank IC is 0.024 ($t = 1.17$), the sector-neutral book returns a net Sharpe of -0.61, and alpha against Fama–French five factors plus momentum is -0.81% ($t = -1.36$). The substantive finding is decay: annual IC falls 0.023 per year ($p = 0.033$), positive through 2022 and negative after — the pattern predicted for an anomaly documented since 1968 and widely traded since the mid-2000s. Run on synthetic data with a known planted effect the same pipeline recovers it in 6/6 years at a net Sharpe of 4.82; with nothing planted it returns 0.022 and 0.37. That contrast is why the null above is worth believing rather than merely asserting.

1 · DESIGN

For each year Y , the model is fitted only on announcements whose 20-day outcome had fully resolved before 1 January Y , less a 25-session embargo; it is then frozen and used to predict every announcement in Y . The target is the cumulative abnormal return from **one session after** the announcement to twenty sessions after: windows beginning on the announcement day contain the opening gap, which cannot be traded, and booking it is the single largest source of overstated drift results. Events come from SEC 8-K Item 2.02 filings, so every one carries a minute-level acceptance timestamp rather than a vendor date — 100% of the sample has a declared, not assumed, before-open or after-close flag. Features are fundamental changes differenced year-on-year and standardised unexpected earnings, each stamped with the moment it became public; a single guard refuses any panel where a feature post-dates the trade. Filing text was not downloaded, so the textual half of the hypothesis is untested here.

2 · SIX HELD-OUT YEARS

Year	Train n	Test n	IC	IC t	Pred. bp	Real. bp	Calib.	Sharpe
2019	1,159	331	0.051	0.77	288	129	0.22	0.91
2020	1,489	338	0.092	2.12	215	228	1.13	1.77
2021	1,829	354	0.027	0.30	299	183	0.37	-1.21
2022	2,181	367	0.045	0.57	256	192	0.56	-0.15
2023	2,548	380	-0.032	-0.45	246	-152	-0.26	-1.48
2024	2,927	391	-0.038	-1.23	230	-26	-0.03	-2.54

Table 1. Each row is a year the model never saw in training. *Pred.* and *Real.* are the predicted and realised top-minus-bottom quintile spreads in basis points; *Calib.* is the slope of realised on predicted, where 1.00 would mean the magnitudes were exactly right. The model predicted a 230–300 bp spread every year and realised between -152 and +228: confident throughout, and wrong more often than not. 2020 is the only year clearing $t = 2$, which is what one year in six looks like under a 5% threshold.

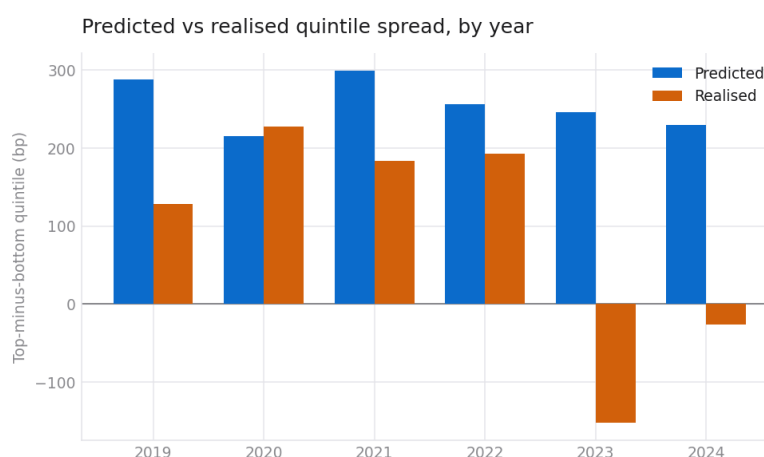


Figure 1. Predicted against realised top-minus-bottom quintile spread, by held-out year. A model with no signal still has opinions.

3 · WHY BELIEVE THE NULL

A null result and a broken pipeline look identical from outside, so the same code was run twice more on synthetic markets where the answer is known in advance. With a post-announcement drift planted in the data it is recovered in 6/6 years at a net Sharpe of 4.82. With nothing planted it returns 0.022 and 0.37 — yet still posts an IC of 0.080 at $t = 3.67$ in one individual year, a false positive in data where there is provably nothing to find. Anyone reporting a single year would have reported that one; the same caution applies to 2020 in Table 1.

Statistic	Real data	Synthetic null
Mean out-of-sample IC	0.024	0.022
t across the six years	1.17	1.15
Years with positive IC	4/6	4/6
Mean realised spread (bp)	92	89
Mean calibration slope	0.33	0.27
Net Sharpe, years stitched	-0.61	0.37
Newey–West t on daily P&L	-1.16	0.70

Table 2. The real study beside the synthetic null control. They look alike, which is the point: the real result is consistent with there being nothing to find.

4 · ALPHA, MULTIPLE TESTING, AND A SECOND METHOD

Regressed on Fama–French five factors plus momentum with Newey–West errors, the book shows an annualised alpha of -0.81% ($t = -1.36$) — no alpha. What the regression does find is significant loadings on value (HML, $t = 2.54$) and investment (CMA, $t = 2.08$): the strategy was unintentionally buying cheap, conservatively-investing firms, which are compensated factors available for a few basis points. Eight specifications are logged, including the two abandoned, giving a deflated Sharpe ratio of 0.14. Fama–MacBeth over 60 monthly cross-sections finds no feature significant at 5%, agreeing with the portfolio sort — which is the outcome that should raise confidence in a null rather than lower it.

5 · THE LIMITATION THAT MATTERS MOST

A point-in-time universe is not enough if the price source cannot serve delisted names. Of 150 tickers sampled at random from every S&P 500 member over the period, 46 had been deleted from the index; Yahoo returned no price history for 28 of those 46 (61%), against 5 of 104 survivors (5%). The names lost include SIVB and FRC — Silicon Valley Bank and First Republic, both of which failed in 2023. Survivorship bias therefore re-enters through the data source even though the universe definition excludes it, and its direction is knowable: the missing firms are disproportionately those that collapsed, so the true result is probably somewhat worse than reported. Separately, SEC XBRL tags quarterly facts inconsistently, so feature coverage runs 10–40% rather than 100%, and filing bodies were not downloaded, leaving the textual half of the hypothesis untested. Fixing the price source needs CRSP or an equivalent with delisting coverage.

6 · REPRODUCIBILITY

make reproduce regenerates every figure and number here and hashes each artefact; make verify diffs against the committed manifest, and CI fails if a hash moves. Method, the falsification criteria fixed before the run (three of five are met), and a catalogue of the ways this study could flatter itself: github.com/pingadom/Earnings-Event-Research.